

POCKET PRODUCER: SESSION-CONDITIONED RETRIEVAL AS USER-CONTROLLED PROPOSAL

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ABSTRACT

We present Pocket Producer, a prototype that operationalizes *session-conditioned continuation retrieval* inside the Audiotool DAW. A five-signal logistic fusion, trained with BPR on pack co-membership weak labels from the Freesound Loop Dataset, serves as the cold-start candidate generator; a production variant of the same recipe is deployed in the system. Four lines of evidence show this proxy should not decide autonomously: (1) the apparent fusion gain is label-sensitive—+4.0 pp over frozen cosine on hard_similar under mixed uploader/pack labels, but −0.03 pp under pack-only labels; (2) leave-one-signal-out ablation reveals regime-dependent signal contributions—tempo helps on hard_similar but hurts on hard_tempo_key; (3) on a pre-committed held-out split the fusion–cosine difference is small and uncertain (hard_similar +1.5 pp, CI crossing zero) and significantly negative on hard_tempo_key; (4) an exploratory pilot shows a descriptive divergence between weak-label signal rankings and human consensus. Pocket Producer therefore treats every recommendation as a *proposal*: candidates are previewable, insertion is reversible, and feedback is logged for future personalization (not yet evaluated).

1. INTRODUCTION

When a musician returns to an unfinished loop-based project, the question is not “what sounds similar?” but “what would be useful as the next step?” Loop compatibility [1–4], stem coherence [5–7], session-conditioned generation [8], and reversible co-creation [9] all address parts of this problem. We define **continuation utility** as the property that a candidate helps the musician make progress in the current session; unlike similarity it is asymmetric and context-dependent.

We contribute: (1) a DAW-integrated prototype whose deployed ranker shares the five-signal fusion recipe evaluated here, treating retrieval as user-controlled proposal, not automatic decision; (2) empirical evidence that weak-label proxy scores are sensitive to label construction, regime-dependent, uncertain on held-out data, and possibly mis-

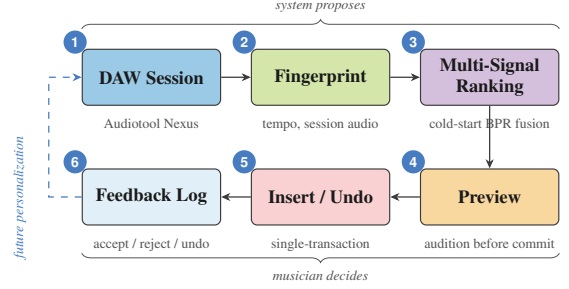


Figure 1. Pocket Producer pipeline: the system proposes (steps 1–3); the musician decides (steps 4–6).

aligned with human judgment—motivating keeping creative authority with the musician.

2. SYSTEM

The live Audiotool session (Figure 1) is read via the Nexus SDK and mapped to a session fingerprint (tempo, and CLAP embeddings [10] of up to eight unique samples referenced by the session’s regions). A five-signal logistic fusion trained with BPR loss [11] ranks candidates across timbral similarity (CLAP cosine to session regions, mean- and max-pooled), tempo match, key compatibility, and tag overlap. The deployed ranker follows the same five-signal recipe but is a separate production model, with four stated deltas from the offline evaluation: it runs in the production MS-CLAP space (offline: LAION-CLAP Music), with weights retrained there on the pack-only train side (mean over 5 seeds); the key term is constant zero online (Audiotool documents carry no key signature); and tag context comes from fragments used in the session rather than pack siblings. It is the default when session audio is available; otherwise the service falls back to a transparent rules baseline. Candidates are surfaced for preview—with per-signal evidence (audio similarity, tempo fit, tag overlap)—before any session state changes. Accepted candidates are inserted via a single Nexus transaction with undo support; rejected or undone events are logged. This design responds to Section 4’s findings: the proxy must not substitute its ranking for the musician’s judgment.

3. WEAK-LABEL FUSION

From the FSLD [12] we kept only sounds with a true Freesound pack (1,797 items, 605 packs), discarding



Table 1. LOSO ablation: accuracy change (pp) when one signal is removed. Pack-only train side (Section 3), mean over 5 splits.

Signal removed	easy	hard_sim.	hard_tk
audio_cos_mean	-0.32	-0.36	-1.04
audio_cos_max	+0.17	-0.30	-0.15
tempo	-0.27	-0.96	+0.68
key	-0.21	+0.94	+1.21
tag_jaccard	+0.09	-0.79	-0.07

items whose only grouping is a shared uploader. Each multi-item pack yields one leave-one-out query per item, with same-pack siblings as positives and cross-pack items as negatives in three difficulty tiers (*easy*: tempo/key mismatch; *hard_tempo_key*: compatible tempo and key; *hard_similar*: highest cosine to the positive from another pack). A 121-source held-out split (seed 20260810) was committed to version control together with the evaluation protocol *before* any evaluation ran. Development uses only the 484 train sources; the 171 with ≥ 3 items form queries (981 examples), the rest contribute negatives only. LAION-CLAP Music [13] (512-dim) provides the embedding backbone.

On the train side (mean over 5 source-grouped splits), frozen cosine reaches 0.929 overall and 0.814 on *hard_similar*; five-signal fusion reaches 0.925 and 0.813. Under pack-only labels the fusion advantage is gone: -0.03 pp on *hard_similar*, and fusion trails cosine on *hard_tempo_key* by 2.1 pp on average (negative in all 5 splits). These development-set numbers are descriptive; inference is reserved for the held-out evaluation (Section 4.3).

4. WHY THE PROXY SHOULD NOT DECIDE ALONE

Four lines of evidence jointly motivate treating the proxy as a proposal rather than a decision.

4.1 The Gain Is Label-Sensitive

An earlier version of this evaluation used the manifest’s fallback grouping, treating *same-uploader* items as co-members in 26% of examples. On that mixed-label corpus, fusion appeared to beat frozen cosine by $+4.0$ pp on *hard_similar* (949 sources, same protocol). Restricting to true pack co-membership erases the gain (Section 3). The restriction changes the source universe and mined negatives together, so the loss cannot be attributed to the uploader-grouped examples alone; what it shows is that the measured advantage is sensitive to label construction—too sensitive to treat as a reliable utility estimate.

4.2 Regime-Dependent Signal Contributions

Table 1 shows leave-one-signal-out deltas per negative regime. Removing tempo costs -0.96 pp on *hard_similar* but *gains* $+0.68$ pp on *hard_tempo_key*; removing key

helps on both hard regimes ($+0.94/+1.21$ pp). Both patterns replicate directionally on the held-out split (tempo $-1.48/+0.37$ pp, key $+0.62/+0.97$ pp). Part of this pattern is built in: *easy* and *hard_tempo_key* are constructed from tempo and key, so those signals are near-constant within them by design. Still, a fused score averages over regimes where signals switch sign, so its ranking cannot be uniformly trusted.

4.3 Held-Out Evaluation

The held-out split was evaluated exactly once, after all development; protocol and split were committed first, making the freeze verifiable in repository history. Of its 121 sources, 54 form queries (323 examples); fusion was retrained on the train side with the same hyperparameters and 5 seeds. The fusion–cosine difference on *hard_similar* is $+1.5$ pp with a source-grouped bootstrap 95% CI of $[-1.5, +4.7]$ pp (joint resampling of the 54 example-bearing source clusters, shared across seeds)—small and uncertain. On *hard_tempo_key*, fusion performs significantly *worse* than cosine (-1.0 pp, CI entirely below zero); overall the difference is $+0.4$ pp with a CI crossing zero.

Residual limitations: the signal set and hyperparameters were designed on the earlier mixed-label corpus, which subsumes this one; the run is a sensitivity re-analysis under a committed split, not *de novo* confirmatory evidence.

4.4 Exploratory Pilot

Nine external raters labeled 20 blind A/B pairs (Fleiss’ $\kappa = 0.294$). One pair whose majority chose “neither” was excluded; 9 of the remaining 19 preference pairs reached $\geq 7/9$ consensus. On these 9 pairs, harmonic compatibility—a signal with no measurable weak-label contribution in our ablations—matches the human majority in 7/9 cases, while tempo, the strongest *hard_similar* contributor in Table 1, matches in 3/9—a descriptive divergence from the weak-label ranking. The comparison is exploratory—post-hoc, small, and on a different corpus and pair construction—raising the possibility of misalignment with human judgment without confirming it.

5. CONCLUSION

Pocket Producer operationalizes session-conditioned retrieval in a DAW. The weak-label fusion is an *operational* cold-start candidate generator—comparable to frozen cosine on held-out data ($+0.4$ pp overall, CI crossing zero; $+1.5$ pp on *hard_similar*, uncertain; worse on *hard_tempo_key*)—that, unlike a bare cosine score, surfaces per-signal evidence with each proposal. Since its contributions switch sign across regimes and may not reflect human preference, it never decides alone: every recommendation is previewable and reversible. Whether preview and undo improve creative outcomes is untested; evaluating that interaction and personalization from logged feedback is future work.

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AI Usage Statement. Claude Code (Anthropic) was used for engineering implementation, experiment scripting, statistical analysis assistance, and documentation drafting. All research claims, experimental design, and conclusions are the sole responsibility of the author.

6. REFERENCES

- [1] B.-Y. Chen, J. B. L. Smith, and Y.-H. Yang, “Neural loop combiner: Neural network models for assessing the compatibility of loops,” in *Proceedings of the 21st International Society for Music Information Retrieval Conference (ISMIR)*, 2020.
- [2] S. Lattner, “SampleMatch: Drum sample retrieval by musical context,” in *Proceedings of the 23rd International Society for Music Information Retrieval Conference (ISMIR)*, 2022.
- [3] X. Yi and M. Barthet, “A generative framework for composition-aware loop recommendation in music production: Drum2Bass use case,” in *Proceedings of the Sound and Music Computing Conference (SMC)*, 2024.
- [4] S. Roy Chaudhuri, S. P. Madupu, K. T. Guru Sai, and V. Jamwal, “LoopMatcher: Proof-of-concept for AI-assisted music loop search,” in *Proceedings of the 15th International Conference on Computational Intelligence in Music, Sound, Art, and Design (EvoMUSART)*, 2026.
- [5] R. Ciranni, G. Mariani, M. Mancusi, E. Postolache, G. Fabbro, E. Rodolà, and L. Cosmo, “COCOLA: Coherence-oriented contrastive learning of musical audio representations,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- [6] J. Nistal, M. Pasini, C. Aouameur, M. Grachten, and S. Lattner, “Diff-A-Riff: Musical accompaniment co-creation via latent diffusion models,” in *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR)*, 2024.
- [7] A. Riou, A. Gagnere, G. Hadjeres, S. Lattner, and G. Peeters, “Zero-shot musical stem retrieval with joint-embedding predictive architectures,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- [8] Q. He, G. Xia, and Z. Wang, “TOMI: Transforming and organizing music ideas for multi-track compositions with full-song structure,” in *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR)*, 2025.
- [9] A. C. Elkins, S. Singh, A. Kieback, S. Blankenship, U. P. Amadasun, and A. Chadha, “DAWZY: A new addition to AI powered “human in the loop” music co-creation,” in *NeurIPS Workshop on AI for Music*, 2025.
- [10] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang, “CLAP: Learning audio concepts from natural language supervision,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [11] S. Rendle, C. Freudenthaler, Z. Gantner, and L. Schmidt-Thieme, “BPR: Bayesian personalized ranking from implicit feedback,” in *Proceedings of the 25th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2009.
- [12] A. Ramires, F. Font, D. Bogdanov, J. B. L. Smith, Y.-H. Yang, J. Ching, B.-Y. Chen, Y.-K. Wu, W.-H. Hsu, and X. Serra, “The freesound loop dataset and annotation tool,” in *Proceedings of the 21st International Society for Music Information Retrieval Conference (ISMIR)*, 2020.
- [13] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, “Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.